

P03 — The Monty Hall Problem

Puzzle Module 03 — Bayes' Theorem & Conditional Probability

Student	Team	Course	Due
Joseph Clay	TuringCollective	ITAI-1370	Jun 20, 2026

1. The Puzzle

You are given three doors to choose from. Behind one door is a car; behind the other two are goats. After you choose a door — but before it is opened — the host, Monty Hall, who knows what is behind every door, opens one of the other two doors to reveal a goat. You are then asked: should you stick with your original choice, or switch to the remaining unopened door?

The additional information *after* the first choice changes the probabilities. The total probability remains one — the car is certainly behind exactly one door.

2. The Answer

You should SWITCH.

Yes — it makes a difference. Switching doubles your probability of winning from 1/3 to 2/3.

3. Justification Using Bayes' Theorem

3.1 The Official Formula

Bayes' Theorem defines how a prior probability is updated when new information is received:

$$P(A | B) = [P(B | A) \times P(A)] / P(B)$$

Where:

$P(A | B)$ = probability the car is behind your door, given Monty revealed a goat

$P(B | A)$ = probability Monty reveals a goat, given the car is behind your door

$P(A)$ = prior probability the car is behind your door = 1/3

$P(B)$ = probability Monty reveals a goat (always happens) = 1

3.2 Step 1 — Initial Probabilities

Before any door is opened, the car is equally likely to be behind any of the three doors:

Door 1	Door 2	Door 3
$P = 1/3$ (33.3%)	$P = 1/3$ (33.3%)	$P = 1/3$ (33.3%)

3.3 Step 2 — Information Gain

Monty's reveal is not random — he always opens a goat door and never your door or the car door. This deliberate action transfers probability. Your door stays locked at $P = 1/3$. The remaining door absorbs the full $2/3$.

3.4 Step 3 — Apply Bayes' Theorem

For the **STAY** scenario:

$$P(car | stay) = [1 \times 1/3] / 1$$
$$P(car | stay) = 1/3 \approx 33.3\%$$

For the **SWITCH** scenario:

$$P(car | switch) = [1 \times 2/3] / 1$$
$$P(car | switch) = 2/3 \approx 66.7\%$$

4. Probability Comparison

Strategy	Initial P	After Reveal	Result
Stay	$1/3$ (33.3%)	$1/3$ (33.3%)	No improvement
Switch	$1/3$ (33.3%)	$2/3$ (66.7%)	Probability DOUBLES

5. Conclusion

The Monty Hall Problem is a classic demonstration of Bayes' Theorem in action. The counterintuitive result — switching wins twice as often as staying — is mathematically proven, not a matter of opinion or intuition.

Because Monty's reveal is deliberate and always shows a goat, it transfers probability to the remaining door. Your original door was selected before any information was available and stays frozen at $P = 1/3$.

Final answer: Always switch. $P(\text{car} \mid \text{switch}) = 2/3$. Yes, it makes a difference — switching doubles your probability of winning the car.

References

Devarakonda, A. (2026). *ITAI-1370 Module 03 — Puzzle 03: The Monty Hall Problem*. Houston Community College.

Mastering the Monty Hall Problem [Video]. (2024). YouTube. <https://youtu.be/CYyUuIXzGgI>

Vos Savant, M. (1990, September 9). Ask Marilyn. *Parade Magazine*. (Original problem statement that made the Monty Hall Problem famous.)

Clay, J. (2026). *Monty Hall interactive simulator — screenshots [Screenshots]*. Claude AI (Anthropic). ITAI-1370, Houston Community College.

Simulator Screenshots

The following screenshots were captured from the Monty Hall interactive simulator built using Claude AI (Anthropic) to demonstrate Bayes' Theorem in real time.

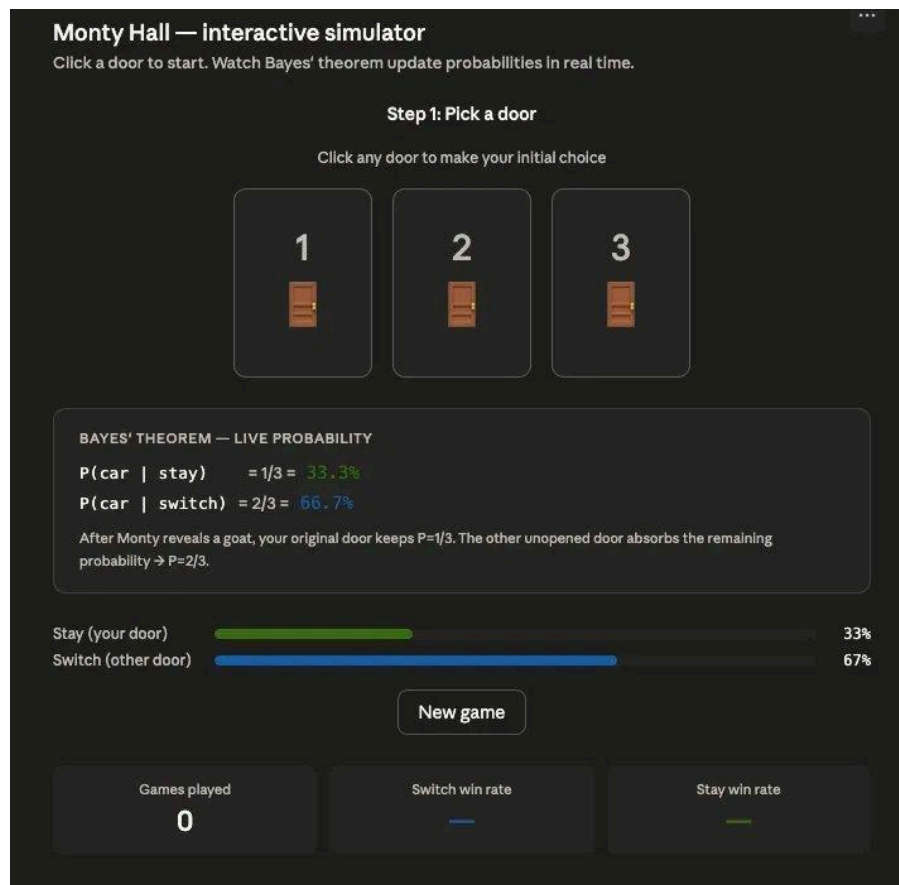


Figure 1 — Step 1: Initial state. Three closed doors, each with $P = 1/3$. Bayes live probability panel shows Stay = 33.3%, Switch = 66.7%.

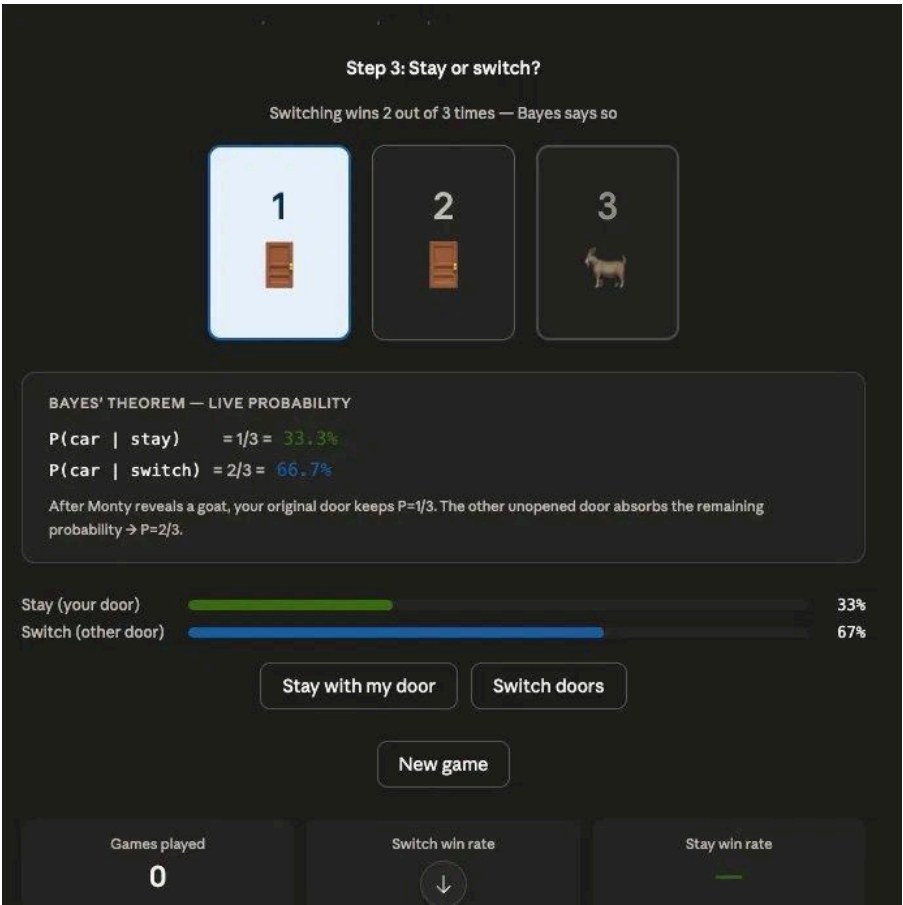


Figure 2 — Step 3: Monty reveals a goat behind Door 3. Decision point: stay with Door 1 or switch to Door 2. Probability bars confirm Switch = 67%.

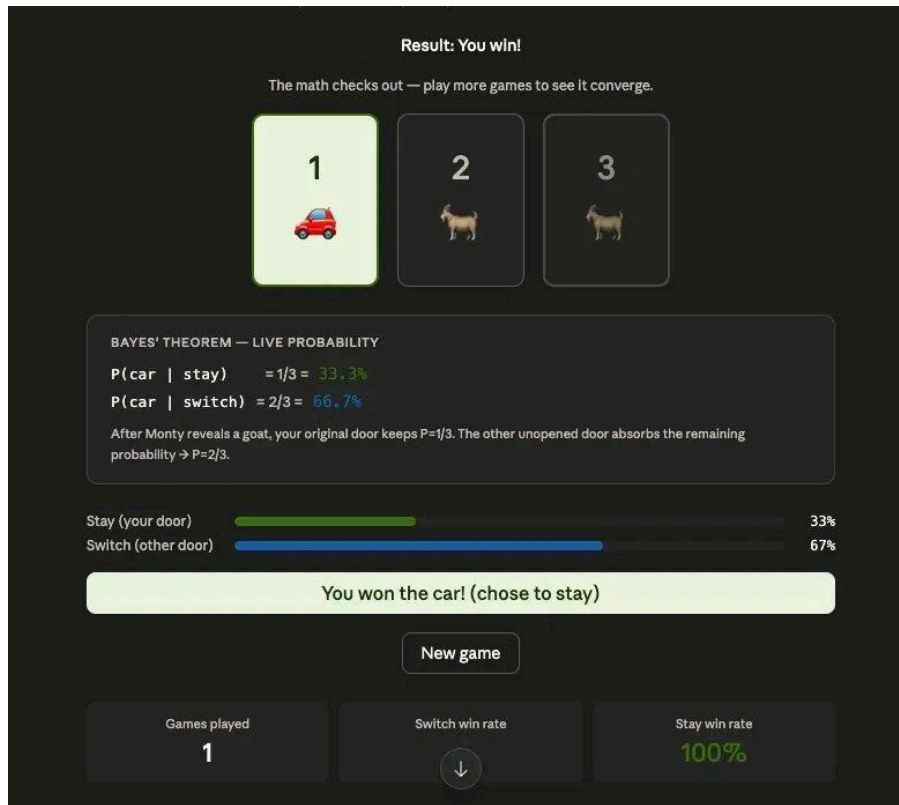


Figure 3 — Result: Car revealed behind Door 1. Player chose to stay and won this round. Stay win rate = 100% after 1 game (small sample); long-run convergence favors switching at 66.7%.